

A Packing of 20 Four-Spaces in $\mathbb{R}^{11}$

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October 9, 2026

Abstract

We give 20 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.678978. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.596364, obtained by deleting 31 of the 51 four-spaces of the packing listed in Cohn’s current table. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 20 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.678978 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 20$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.596364, obtained by deleting 31 of the 51 four-spaces of the packing listed in Cohn’s current table. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 20/19 = 2.679426$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N20.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 68 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
267897839699228946949108721784737291685330485807326293633718953238270275
383000350760963447424585452692842590855260081441947820693272326174793043
6668899247147111333733482541216966470976177554709
```

denominator:

```
100000000000107048700656687561279149538960818225731267763505405687763267
635233744975941117662844226845945978365670474454500273331946529090156735
2099157227608996385688411127870097178710184783432
```

B Coordinates

20 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N20.txt`.

```
-330743828408 460074217500 358571400082 3436698749207514608327 187296397861 269097708229 -44527675736 34110523953 -225440200475 203397054237 -302995212044
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